3. **Assertion:** The number of protons or the number of neutrons is known as atomic number.
   Reason: The mass number is the sum of protons and neutrons.

VII. Answer very briefly.

1. Define – Atom.
2. Name the sub-atomic particles.
3. What is atomic number?
4. What are the characteristics of proton?
5. Why neutrons are called neutral particles?

VIII. Answer briefly.

1. Distinguish isotopes from isobar.
2. What are isotones? Give one example.
3. Differentiate mass number from atomic number.
4. The atomic number of an element is 9. It has 10 neutrons. Find the element from the periodic table. What will be its mass number?

IX. Answer in detail.

1. Draw the structure of an atom and explain the position of the sub-atomic particles.
2. The atomic number and the mass number of an element is 26 and 56 respectively. Calculate the number of electrons, protons and neutrons in its atom. Draw the structure.
3. What are nucleons? Why are they called so? Write the properties of the nucleons.
4. Define valency. What is the valency of the element with atomic number 8? What is the compound format by this element with hydrogen?

X. Higher Order Thinking Skills.

1. An atom of an element has no electron. Will that atom have any mass or not? Can an atom exist without electron? If so then give example.
2. What is common salt? Name the elements present in it. Write the formula of common salt. What are the atomic number and the mass number of the elements? Write the ions in the compound.

XI. Project.

To have an idea of what atoms are, students can be asked to construct atoms using pipe cleaners (thin metal wires-electron shells), pom-poms (balls-different colours for protons and neutrons) and beads (electrons). Students will love and enjoy putting them together and they look great hanging from the ceiling in the classroom.

VII Std Science Term-1 EM Unit 4.indd 52 24-03-2020 14:21:32